Comptia N+

TROUBLESHOOTING AND HELPDESK

DAY2

TASKS:1

List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

RESULT

Hardware Issue	**Typical Symptom**
No display on monitor	Blank screen, no POST beep.
Faulty RAM	PC beeps continuously or restarts repeatedly.
Hard disk failure	Operating system does not boot or "No Boot Device" error appears.
Faulty Power Supply (SMPS)	PC does not turn on or shuts down unexpectedly.
CPU Overheating	PC becomes very hot and shuts down automatically.

TASKS:2

Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools

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(no software diagnostics), and explain what you would check at each step.

RESULT

Step 1: Check the Power Outlet

- Plug another device into the same socket.
- **Check:** Is the power outlet working?

Step 2: Inspect the Charger

- Look for cuts, burns, or loose connections.
- **Check:** Is the charger physically damaged?

Step 3: Check the Charging Port

- Inspect the laptop charging port.
- **Check:** Is the port loose, dirty, or damaged?

Step 4: Check the Charging Indicator

- Connect the charger.
- **Check:** Does the charging LED turn on?

Step 5: Test with Another Charger

- Use another compatible charger.
- **Check:** Does the laptop start charging?

Step 6: Check the Battery

- If possible, remove and reconnect the battery (for removable batteries).

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- **Check:** Is the battery properly connected?

TASKS:3

Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

RESULT

1. Set the multimeter to **DC Voltage (20V range)**.
2. Plug the USB phone charger into a power outlet.
3. Identify the **+5V** and **Ground (GND)** pins of the USB connector.
4. Place the **black probe** of the multimeter on the **Ground (GND)** pin.
5. Place the **red probe** on the **+5V** pin.
6. Read the voltage displayed on the multimeter.
7. Record the measured voltage.
8. Compare the measured value with the standard USB voltage range

TASKS:4

A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the

multimeter to DC voltage and probe between the yellow and black wires.</em\

RESULT

- Turn off the computer and disconnect the power cable.
- Disconnect the **24-pin ATX power connector** from the motherboard.
- Set the multimeter to **DC Voltage (20V range)**.
- Locate a **yellow wire (+12V)** and a **black wire (Ground)** on the ATX connector.
- Connect the **black probe** to the **black (Ground)** wire.
- Connect the **red probe** to the **yellow (+12V)** wire.
- Turn on the power supply .
- Read the voltage displayed on the multimeter.
- Record the measured voltage.
- Compare the measured value with the expected **12V** output.